# Python Project

## Importing necessary librarys

In [3]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

## Preprocessing:

Correct the data in the "height" column by replacing it with random numbers between 150 and 180. Ensure data consistency and integrity before proceeding with analysis

In [5]:
```
Teams = pd.read_excel("myexcel.xlsx")                                            ## Importing
Teams['Salary'] = Teams['Salary'].fillna('0')                                    ## replaces
Teams['Salary'] = Teams['Salary'].apply(int)                                     ## converts
Teams['Salary'] = Teams['Salary'].replace(0,(int(Teams['Salary'].mean())))       ## Replacing
Teams = Teams.fillna('')                                                         ## replace N
Teams = Teams.drop_duplicates()                                                  ## removes a
Teams["Height"] = np.random.randint(150,160+1,len(Teams))                        ## replacing
Teams.head(5)
```

Out[5]:

| | Name | Team | Number | Position | Age | Height | Weight | College | Salary |
|---|---|---|---|---|---|---|---|---|---|
| **0** | Avery Bradley | Boston Celtics | 0 | PG | 25 | 159 | 180 | Texas | 7730337 |
| **1** | Jae Crowder | Boston Celtics | 99 | SF | 25 | 152 | 235 | Marquette | 6796117 |
| **2** | John Holland | Boston Celtics | 30 | SG | 27 | 156 | 205 | Boston University | 4717869 |
| **3** | R.J. Hunter | Boston Celtics | 28 | SG | 22 | 150 | 185 | Georgia State | 1148640 |
| **4** | Jonas Jerebko | Boston Celtics | 8 | PF | 29 | 159 | 231 | | 5000000 |

## Analysis Tasks:

## 1.Determine the distribution of employees across each team and calculate the percentage split relative to the total number of employees

In [8]:
```
team_counts = Teams['Team'].value_counts()                          # Group by tea
total_els = len(Teams)                                              # Calculate to
percentage_split = (team_counts/total_els) * 100                    # Calculate Pe

## Create a new DataFrame with team, count, and percentage ##
team_distribution = pd.DataFrame({'Team': team_counts.index,
                                  'Count': team_counts.values,
                                  'Percentage': percentage_split.values})
team_distribution.head(3)
```

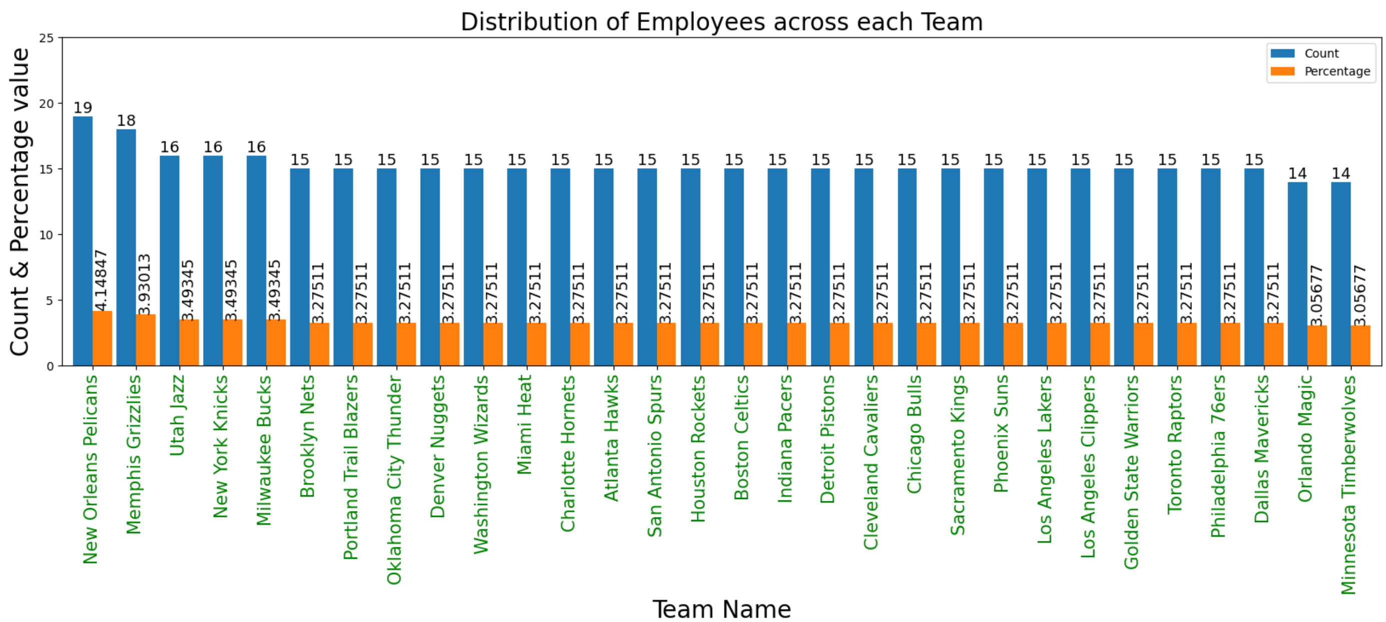
Out[8]:

| | Team | Count | Percentage |
|---|---|---|---|
| **0** | New Orleans Pelicans | 19 | 4.148472 |
| **1** | Memphis Grizzlies | 18 | 3.930131 |

| | | | |
|---|---|---|---|
| **2** | Utah Jazz | 16 | 3.493450 |

### Ploting for distribution of employees across each team

In [10]:
```
ax=team_distribution.plot.bar(x='Team',stacked=False,width=0.9,figsize=(20,5))
plt.ylim(top=25)
plt.ylim(bottom=0)
plt.title("Distribution of Employees across each Team",size = 20)
plt.xticks(size=15,color ='g')
plt.xlabel('Team Name',size = 20)
plt.ylabel('Count & Percentage value',size = 20)
ax.bar_label(ax.containers[0], label_type='edge',size = 13)
ax.bar_label(ax.containers[1], label_type='edge',rotation=90,size = 13)
plt.show()
```




## 2. Segregate employees based on their positions within the company

In [12]:
```
# Segregate employees based on their positions
grouped_Teams = Teams.groupby(['Position']).agg({'Name': 'nunique','Team':'nunique',})
grouped_Teams
```

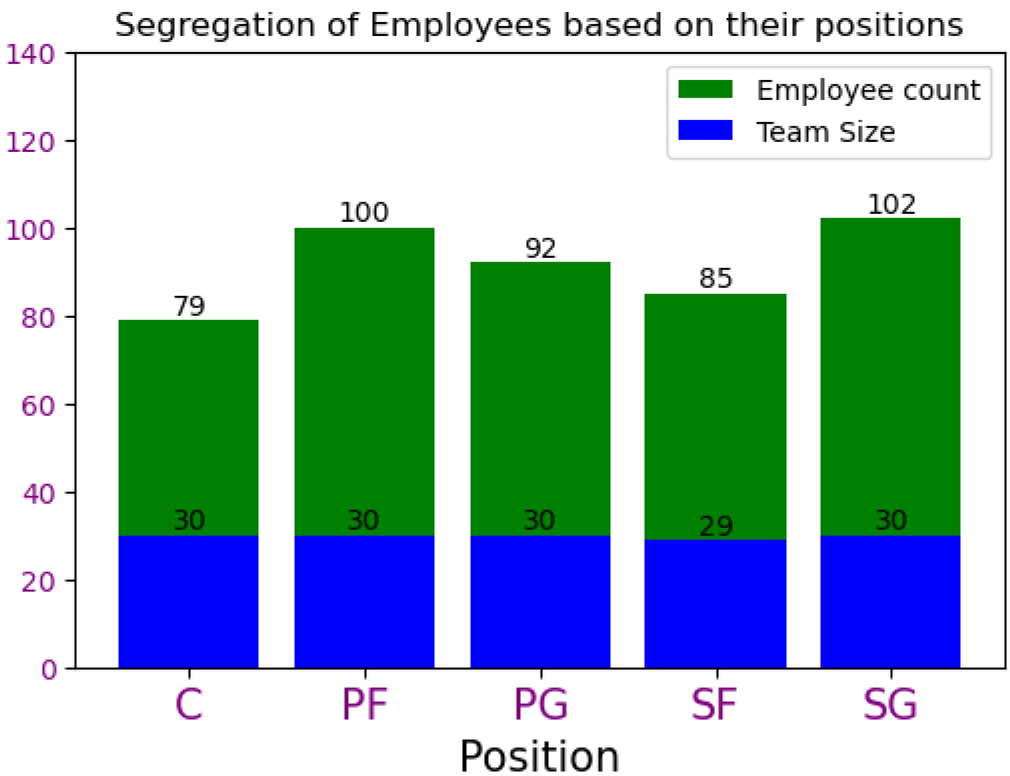
Out[12]:

| | Name | Team |
|---|---|---|
| **Position** | | |
| **C** | 79 | 30 |
| **PF** | 100 | 30 |
| **PG** | 92 | 30 |
| **SF** | 85 | 29 |
| **SG** | 102 | 30 |

### Ploting Segregation of employees based on their positions

In [14]:
```
plt.ylim(top=140)
plt.ylim(bottom=0)
ax=grouped_Teams['Name'].plot.bar(stacked=True,figsize=(6,4),color ="g",label='Employee
ax=grouped_Teams['Team'].plot.bar(stacked=True,figsize=(6,4),color ="b",label='Team Size
plt.xticks(rotation=360,size=15,color ='purple')
```

```
plt.yticks(size=10,color ='purple')
ax.bar_label(ax.containers[0], label_type='edge')
ax.bar_label(ax.containers[1], label_type='edge')
plt.title("Segregation of Employees based on their positions",size = 12)
plt.xlabel('Position',size = 15)
plt.grid(False)
plt.show()
```




## 3. Identify the predominant age group among employees.

In [16]:
```
grouped_Age = Teams.groupby(['Age']).agg({'Name': 'nunique'})
grouped_Age.head(5)
```

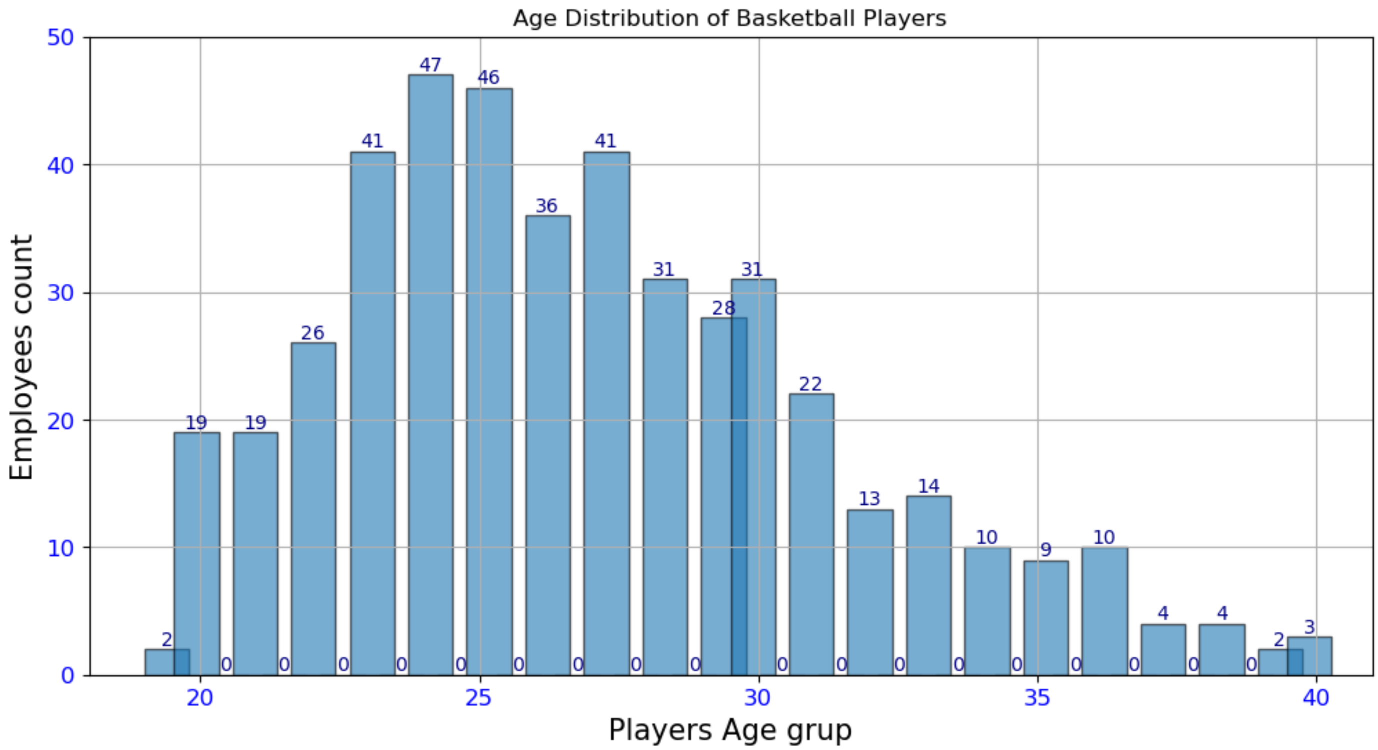
Out[16]:

| | Name |
|---|---|
| **Age** | |
| **19** | 2 |
| **20** | 19 |
| **21** | 19 |
| **22** | 26 |
| **23** | 41 |

## Ploting Age Group among Employees

In [18]:
```
plt.figure(figsize=(12, 6))
plt.ylim(top=50)
plt.ylim(bottom=0)
plt.xlim(18,41)
ax=plt.hist(Teams['Age'], bins=40,edgecolor='black',width=.8,alpha=.6,bottom=None)
plt.bar_label(ax[2],fontsize=10, color='navy' )
plt.xticks(rotation=360,size=12,color ='b')
plt.yticks(size=12,color ='b')
```

```
plt.xlabel('Players Age grup',size = 15)
plt.ylabel('Employees count',size = 15)
plt.title('Age Distribution of Basketball Players')
plt.grid(True)
plt.show()
```




## 4. Discover which team and position have the highest salary expenditure

In [20]:
```
grouped_df = Teams.groupby(['Team'])['Salary'].sum()
Hig_exp_tem = grouped_df.idxmax()
Hig_exp_amount = grouped_df.max()
print(f'\033[1mHighest expenditure team is "{Hig_exp_tem}" and Hig expenditure Amount is
grouped_df = pd.DataFrame(grouped_df)
grouped_df.head(5)
```

**Highest expenditure team is "Cleveland Cavaliers" and Hig expenditure Amount is "111706558"**

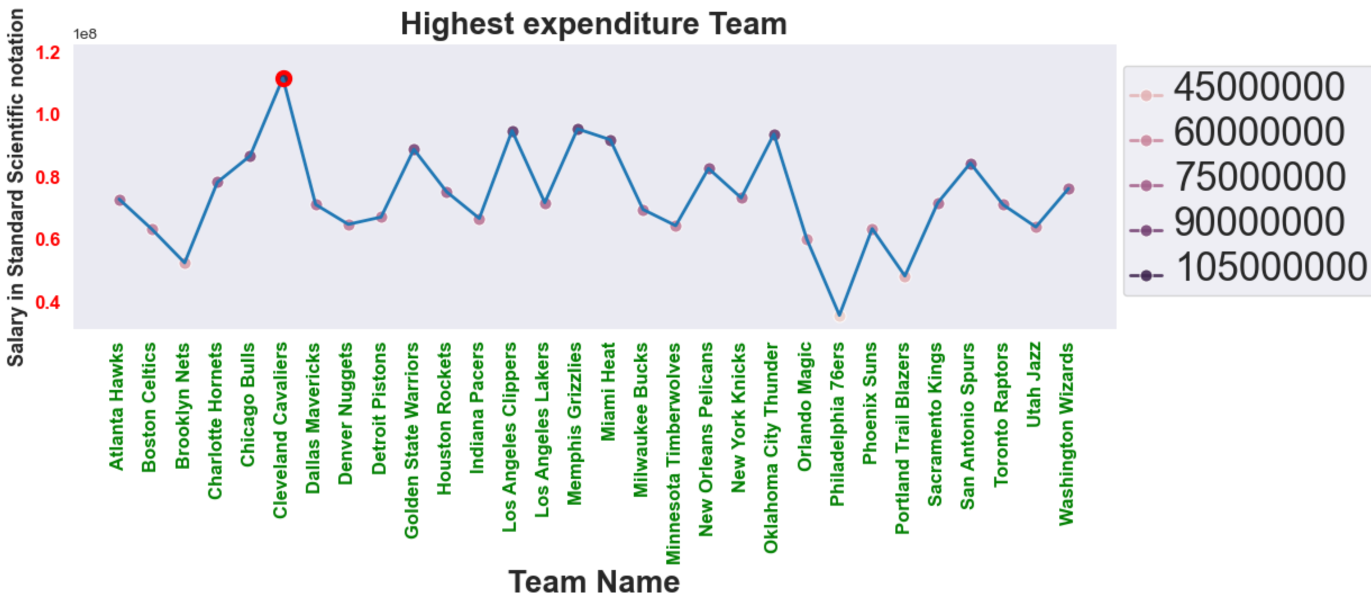
Out[20]:

| Team | Salary |
|---|---|
| **Atlanta Hawks** | 72902950 |
| **Boston Celtics** | 63258937 |
| **Brooklyn Nets** | 52528475 |
| **Charlotte Hornets** | 78340920 |
| **Chicago Bulls** | 86783378 |

## Ploting Highest expenditure Team

In [22]:
```
sns.set_style('dark')
plt.figure(figsize=(12, 3.3))
plt.title('Highest expenditur team')
lpts=sns.lineplot(data= grouped_df, x='Team', y='Salary', marker='o', markersize=8,hue='
sns.lineplot(data= grouped_df, x='Team', y='Salary',linewidth=2)
lpts.plot(Hig_exp_tem, Hig_exp_amount, marker= "$\\circ$", color="r", markersize=10)
```

```
lpts.set_ylim(top=122706558)
lpts.set_ylim(bottom=31100000)
lpts.set_ylabel("Salary in Standard Scientific notation",fontsize=13,weight='bold')
lpts.set_xlabel("Team Name",fontsize=20,weight='bold')
lpts.set_title('Highest expenditure Team',fontsize=20,weight='bold')
sns.move_legend(lpts, "upper left", bbox_to_anchor=(1,.9509))
plt.setp(lpts.get_legend().get_texts(), fontsize='25') # for legend text
plt.setp(lpts.get_legend().get_title(), fontsize='32') # for legend title
plt.xticks(size=12,weight='bold',color ='green',rotation = 90)
plt.yticks(size=12,weight='bold',color ='red')
plt.show()
```




## 5. Investigate if there's any correlation between age and salary, and represent it visually

In [24]:

```
# Computing the correlation between Age and Salary
corr_Teams =Teams[['Age','Salary']]
correlation = corr_Teams.corr()
correlation
```

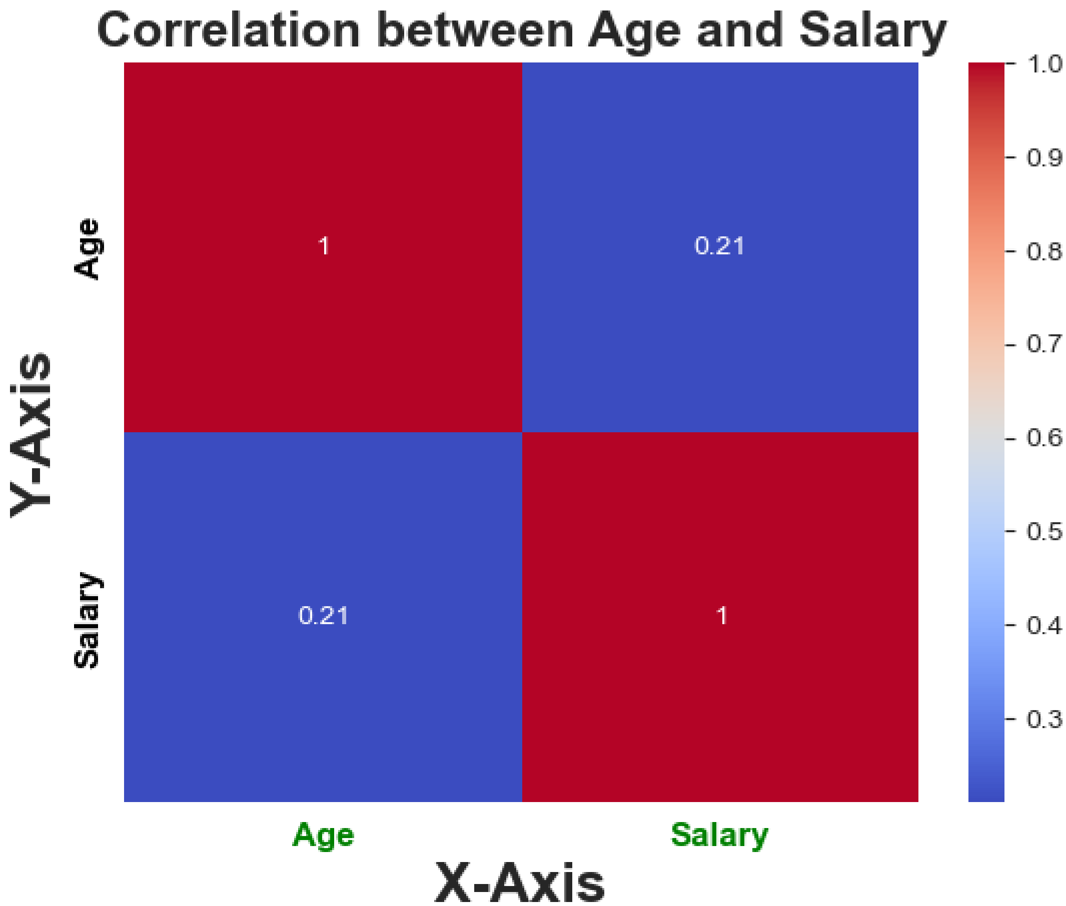
Out[24]:

| | Age | Salary |
|---|---|---|
| **Age** | 1.000000 | 0.211071 |
| **Salary** | 0.211071 | 1.000000 |

## Ploting Heat map for correlation

In [26]:

```
heatm=sns.heatmap(correlation,annot=True, cmap='coolwarm')
heatm.set_ylabel("Y-Axis",fontsize=20,weight='bold')
heatm.set_xlabel("X-Axis",fontsize=20,weight='bold')
heatm.set_title('Correlation between Age and Salary',fontsize=18,weight='bold')
plt.xticks(size=12,weight='bold',color ='green')
plt.yticks(size=12,weight='bold',color ='black')
plt.show()
```



Correlation between Age and Salary
Y-Axis
Age
Salary
1
0.21
0.21
1
Age
Salary
X-Axis
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3